## Host and burst properties vs axis ratio: outlier-robust statistics

| Property vs axis ratio $q$ | N | $r_S$ (p) | AD p | bootstrap 95% | Kendall $\tau$ (p) | delete-1 range | Theil–Sen slope [95%] |
|---|---|---|---|---|---|---|---|
| $DM_{host}$ (rest frame) | 55 | -0.06 (0.64) | 0.37 | [-0.32, +0.20] | -0.04 (0.63) | -0.12 to -0.04 | -38 [-239, 159] |
| $DM_{host}$, foreground halos removed | 49 | -0.04 (0.76) | 0.68 | [-0.33, +0.25] | -0.03 (0.74) | -0.10 to +0.00 | -51 [-288, 165] |
| $DM_{host}$, foreground excess over random sightlines removed | 49 | -0.08 (0.60) | 0.65 | [-0.35, +0.21] | -0.06 (0.56) | -0.13 to -0.03 | -67.6 [-278, 147] |
| $DM_{host}$, foreground removed (all photo-$z$ galaxies in front) | 49 | -0.07 (0.62) | 0.90 | [-0.35, +0.22] | -0.04 (0.67) | -0.12 to -0.02 | -42.6 [-292, 178] |
| $DM_{host}$, spectroscopic foreground only | 49 | -0.06 (0.68) | 0.41 | [-0.34, +0.23] | -0.04 (0.72) | -0.12 to -0.02 | -32.6 [-243, 147] |
| log\|RM\| | 41 | -0.02 (0.91) | 0.35 | [-0.31, +0.30] | -0.00 (0.99) | -0.09 to +0.04 | -0.00544 [-1.46, 1.12] |
| $\log \tau_{scatt}$ | 29 | -0.17 (0.39) | 0.64 | [-0.55, +0.27] | -0.11 (0.41) | -0.28 to -0.07 | -1.03 [-2.92, 1.48] |
| Linear polarization | 24 | -0.33 (0.11) | 0.50 | [-0.69, +0.12] | -0.24 (0.096) | -0.45 to -0.25 | -0.467 [-1.08, 0.188] |
| $A_V$, old stars | 30 | -0.46 (0.010) | 0.003 | [-0.67, -0.15] | -0.28 (0.032) | -0.53 to -0.41 | -0.429 [-1.06, -0.0207] |
| $DM_{host}$ (rest frame), m_r < 21 | 45 | -0.09 (0.54) | 0.080 | [-0.38, +0.19] | -0.07 (0.52) | -0.16 to -0.06 | -54 [-273, 168] |
| $DM_{host}$ (rest frame), n < 2 | 46 | -0.04 (0.81) | 0.59 | [-0.32, +0.26] | -0.02 (0.85) | -0.10 to +0.00 | -10.5 [-177, 168] |
| $DM_{host}$ (rest frame), bulge fits removed | 52 | -0.07 (0.64) | 0.25 | [-0.34, +0.20] | -0.04 (0.64) | -0.12 to -0.04 | -38.4 [-254, 168] |
| $DM_{host}$, foreground halos removed, m_r < 21 | 39 | -0.04 (0.81) | 0.47 | [-0.36, +0.28] | -0.03 (0.76) | -0.11 to +0.02 | -41.5 [-256, 173] |
| $DM_{host}$, foreground halos removed, n < 2 | 42 | -0.00 (0.99) | 0.48 | [-0.31, +0.30] | -0.00 (0.99) | -0.06 to +0.05 | -0.495 [-274, 204] |
| $DM_{host}$, foreground halos removed, bulge fits removed | 46 | -0.04 (0.79) | 0.58 | [-0.33, +0.26] | -0.03 (0.77) | -0.10 to +0.01 | -46.9 [-290, 167] |

Bootstrap: 10,000 resamples. Since every host has $q > q_0$, rank correlations with $q$ and with $\cos i$ are identical (sign flipped for $i$).
AD p: two-sample Anderson–Darling test that y (in 'y vs x'; the property, against $q$) has the same distribution for hosts below and above the median of x (permutation p, about 5,000 shuffles).
$A_V$ of the old stars is the only Spearman p < 0.05 case (dust in inclined disks).